[PS-5] RATE BASED CONGESTION CONTROL

GitHub : Link

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Problem Description

- Two hosts (Host 1 and Host 2) share a single bottleneck network link, leading to congestion when both send aggressively.
- Each host independently selects a sending rate from three options: **Low, Medium, High.**
- The interaction is modeled as a **3 × 3 strategic-form game**, where each pair of actions decides the outcome.
- Each host's payoff is defined as: **Payoff = Throughput − Congestion Cost**, where congestion cost increases with the total sending rate of both hosts.

Problem Objectives

- Construct the payoff matrix for all action pairs
- Find **Pure Strategy Nash Equilibria (PSNE)** and **Mixed Strategy Nash Equilibrium (MSNE)**
- The game captures the trade-off between:
  - **Higher individual throughput (favoring High rate)**
  - **Increased congestion (penalizing total rate)**
- A simulation of repeated interactions is performed to:
  - Model dynamic behavior of hosts over time
  - Verify that the empirical average payoff converges to the theoretical equilibrium payoff

Game Theoretic Modelling (Strategy Form)

- $N = 2$, $S_1 = S_2 = \{Low, Mid, High\}$
- $U_1(x,y) = x - a(x+y)$, $U_2(x,y) = y - a(x+y)$
- Mapping of x/y : {Low:1, Med:2, High:3}
- Here, alpha is a parameter that shows the penalizing effect of congestion on individual utility.
- Special Case: a = 0.5, here the game becomes a zero-sum game thus we can find MSNE using LP optimization.

Utility Matrix (a = 0.5)	**L**	**M**	**H**
L	(0, 0)	(-0.5, 0.5)	(-1, 1)
M	(0.5, -0.5)	(0, 0)	(-0.5, 0.5)
H	(1, -1)	(0.5, -0.5)	(0, 0)

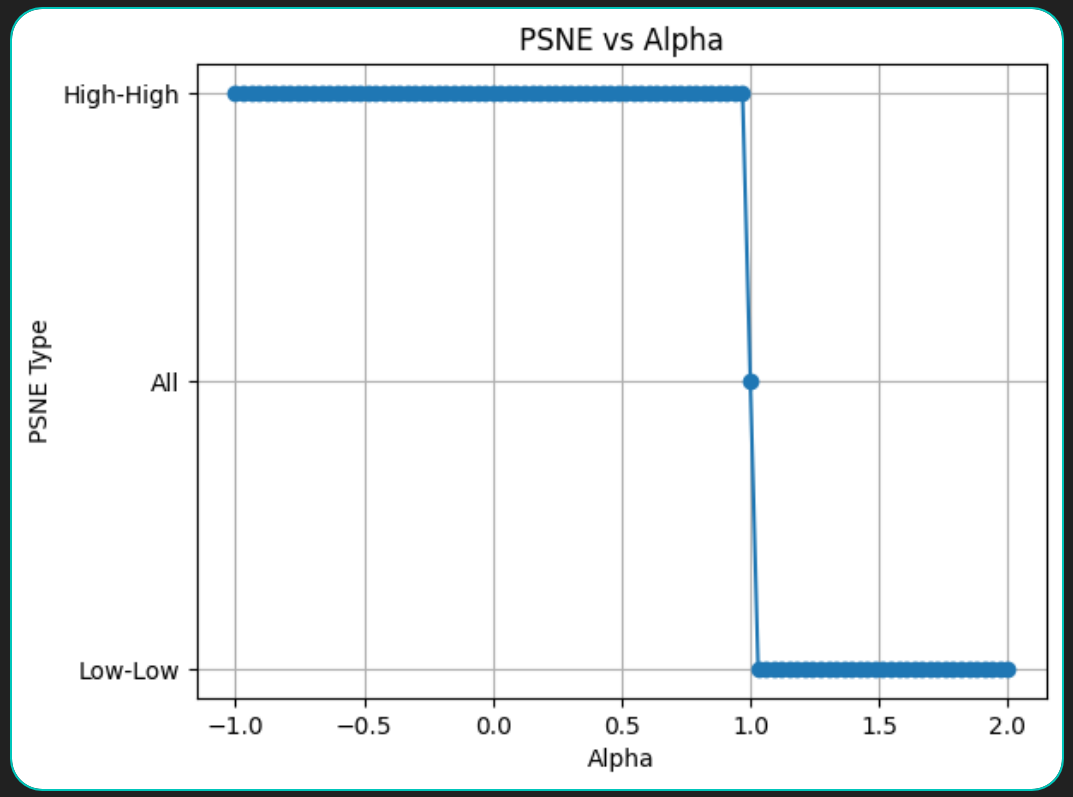
Methodology for analysis

- Identification of **Pure Strategy Nash Equilibria (PSNE)**:
  - We compute the best response for each player.
  - For **a = 0.5**, we find saddle point and verified that PSNE from best response is equal to the saddle point.
- Identification of **Mixed Strategy Nash Equilibria (MSNE)** using LP:
  - When **a = 0.5**, we used **LP** to find MSNE as the game is 0 sum game.
  - Our aim was to max Z, st
    - $Z - (A_{11}X_1 + A_{21}X_2 \text{---------} A_{n1}X_n) <= 0$
    - $Z - (A_{12}X_1 + A_{22}X_2 \text{---------} A_{n2}X_n) <= 0$
    - $Z - (A_{1m}X_1 + A_{2m}X_1 \text{----------} A_{nm}X_n) <= 0$
    - $\sum X_i = 1$
    - $X_i >= 0$

Methodology for analysis

- Identification of **Mixed Strategy Nash Equilibria (MSNE)** using Indifference Method:
  - $E[U_1(s_1)] = E[U_1(s_2)] = \cdots = E[U_1(s_k)]$
  - $E[U_2(s_1)] = E[U_2(s_2)] = \cdots = E[U_2(s_k)]$
- Simulation for Validation:
  - We use **python** to simulate repeated interaction between the two hosts.
  - Procedure:
    - Use **MSNE** probabilities (**p***, **q***) obtained above.
    - For each iteration:
      - Sample strategies randomly according to these probabilities.
      - Compute payoff using utility matrices.
    - Repeat for a large number of interactions

PSNE vs a ε [-1,2]



- Fixing opponent's action as c: U(x,c) = x(1−a)−ac, so only 1−a decides the best response.
- **If a > 1:** 1−a<0 ⇒ utility decreases for x ⇒ best response is **Low** ⇒ **PSNE: (Low,Low)**.
- **If a<1:** 1−a>0 ⇒ utility increases in x ⇒ best response is **High** ⇒ **PSNE: (High,High)**.
- **If a=1:** utility independent of x ⇒ all strategies yield same payoff ⇒ **all profiles are PSNE**.

MSNE (a=0.5)

- LP formulation:
  - Max Z:
    - $Z - (0*X_1 + 0.5*X_2 + 1*X_3) <= 0$
    - $Z - (-0.5*X_1 + 0*X_2 + 0.5*X_3) <= 0$
    - $Z - (-1*X_1 + -0.5*X_2 + 0*X_3) <= 0$
    - $X_1 + X_2 + X_3 = 1$
    - $X_1 >=0, X_2 >=0, X_3 >=0$
  - Using above formulation MSNE was coming out to be:
    - For player 1: (0,0,1)
    - For player 2: (0,0,1)

MSNE (using indifference methods)

$a < 1$

- Result: MSNE converges to **(High, High)**.
- His utility is $(1- a)x - ay$, and $1-a > 0$, so he tries to max x.

$a > 1$

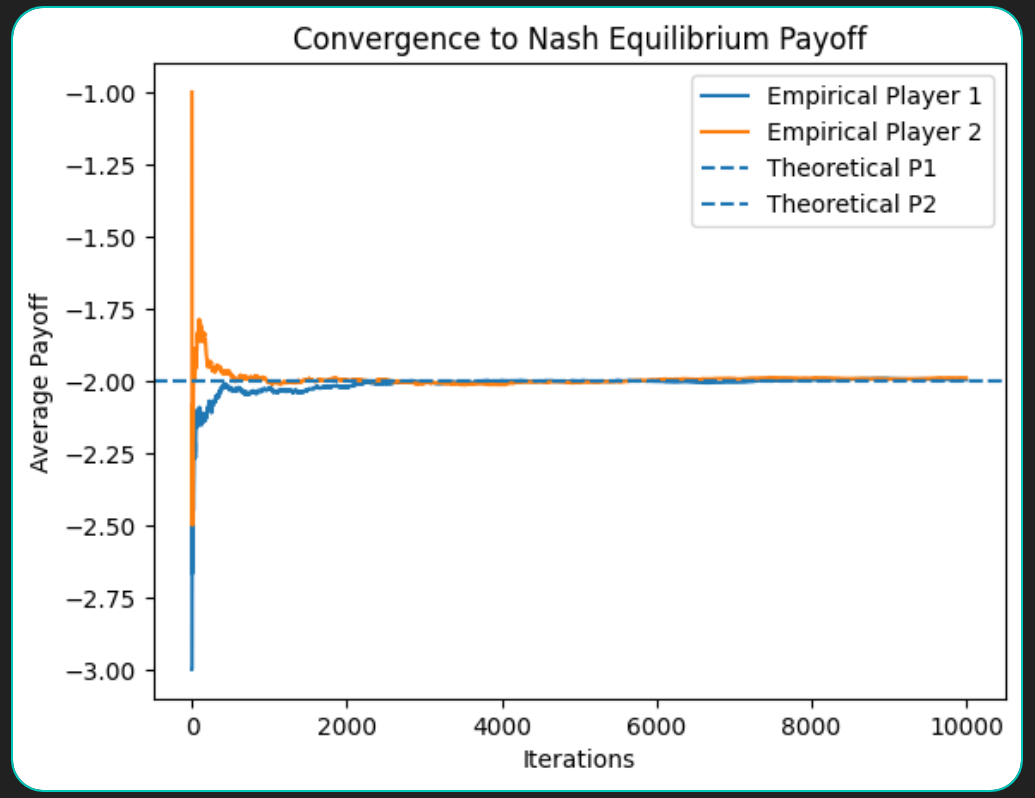
- Result: MSNE converges to (Low, Low)
- His $(1-a) < 0$ so he tries to minimize x.

$a = 1$

- At $a=1$: $U_1=-y$, $U_2=-x$
- Result: Infinite MSNE possibilities.
- His $(1-a) = 0$ so his actions does not depend on his utility.

MSNE for alpha = 1 simulation

- For S_1 = (0.33,0.33,0.33) , S_2= (0.33, 0.33, 0.33)
- Expected Utility = 0.33*0.33*x1y1 + ... = -2
- We can see that average utility from simulations converges to Expected Utility over iterations.



Inferences

- Hosts act **selfishly**, creating a trade-off between **throughput** and **congestion cost**
- **Low a (weak penalty):** both choose High → **network overload risk**
- **High a (strong penalty):** both choose Low → **underutilization**
- Outcome depends critically on **congestion sensitivity (a)**
- **Selfish behavior ≠ optimal network performance**
- Highlights need for **congestion control mechanisms** to ensure stability and efficiency

Learning outcome

- Understand how **user behavior in networks can be modeled using game theory**
- Learn that **selfish rate choice can lead to inefficiency** (overload or underutilization)
- Recognize the importance of **balancing throughput and congestion cost**
- Apply this insight to design **congestion control mechanisms or protocols** that:
  - Regulate sending rates
  - Ensure fair bandwidth sharing
  - Maintain network stability

Thank you!

Work Division

Simulations

- Siddharth
- Krish

Slides

- Shreyansh